

Engineering Physics (19PH102)

(Review Questions for MSE & SEE)

UNIT I: Wave Mechanics

1. What are matter waves? Mention their characteristics. (3 or 6 marks)
2. What is de Broglie concept of matter waves? Derive an expression for de Broglie wavelength. (3 or 6 marks)
3. What are matter waves? Write a note on de Broglie hypothesis of matter waves. (3 or 6 marks)
4. Define phase velocity. Obtain an expression for the same. (3 or 6 marks)
5. Define group velocity. Obtain an expression for the same. (3 or 6 marks)
6. Discuss phase velocity and group velocity. (3 or 6 marks)
7. Write a note on phase velocity and group velocity. (3 or 6 marks)
8. Derive an expression for group velocity on the basis of superposition of two travelling waves. (3 or 6 marks)
9. State and explain Heisenberg's uncertainty principle. (3 or 6 marks)
10. What is a wave function? Mention important characteristics of wave function. (3 or 6 marks)
11. Discuss matter waves and wave function. (3 or 6 marks)
12. Write a note on matter waves and wave function. (3 or 6 marks)
13. State and explain Heisenberg's uncertainty principle. (3 or 6 marks)
14. What is a wave function? Give its physical significance. What is normalization of a wave function? (3 or 6 marks)
15. Define the terms a) Wave function, b) Eigen value and c) Eigen function. (3 or 6 marks)
16. What is a wave function? Discuss its physical significance. (3 or 6 marks)
17. Discuss probability density and normalization of a wave function. (3 or 6 marks)
18. Explain the terms a) Phase velocity b) Group velocity and c) Free particle. (3 or 6 marks)
19. Explain the following terms: (i) Probability density (ii) Normalization of a wave function (3 or 6 marks)
20. Derive Schrodinger's time independent one-dimensional wave equation for a particle of mass m with energy E . (4 or 10 marks)
21. Set up the time independent Schrodinger's wave equation for a particle in one dimension. (4 or 10 marks)
22. What is a free particle? Solve Schrodinger's wave equation for a free particle. What conclusions can we draw about the energy of the free particle? (4 or 6 marks)
23. Solve the Schrodinger's wave equation for a particle in one dimensional potential well of infinite height and discuss about energy Eigen values. (4 or 10 marks)

24. Solve Schrodinger's wave equation for a particle in an infinitely deep potential well of width L and show that the energy values are quantised. (4 or 10 marks)
25. Obtain the solution of Schrodinger's wave equation for a particle in one dimensional potential well of infinite height and finite width. (4 or 10 marks)
26. Obtain the expression for energy Eigen value and Eigen function for a particle in a one dimensional potential well of infinite height. (4 or 10 marks)
27. Assuming the solution of Schrodinger's wave equation for a particle in an infinitely deep potential well, plot the wave function and probability density as a function of position inside the well. What conclusions can we draw from them? (4 or 10 marks)
28. Discuss the wave function, probability density and energy Eigen values for a particle in a potential well by considering its ground state and the first two excited states. (4 or 10 marks)
29. Using Schrodinger's wave equation for a particle in one dimensional potential well of infinite height discuss wave functions, energy levels and probability densities. (4 or 10 marks).

Problems: (3 or 4 marks)

1. Find the de Broglie wavelength associated with an electron travelling with a velocity 10^6 m/s.
2. Calculate the de Broglie wavelength associated with an electron with a kinetic energy of 2 keV.
3. Calculate the de Broglie wavelength associated with an electron subjected to a potential difference of 1.25 kV.
4. An electron beam is subjected to a potential of 10^3 volts. Find the de Broglie wavelength associated with the electrons.
5. Compare the energy of a photon with that of an electron when both are associated with a wave length of $1\text{\AA}$.
6. An electron is bound in a one-dimensional potential well of width $1\text{\AA}$, but of infinite wall height. Find its energy values in the ground state and in the first two excited states.
7. An electron is bound in a one-dimensional potential well of width $4\text{\AA}$, but of infinite wall height. Find its first three energy values.
8. An electron is trapped in a one-dimensional region of length $1.2\text{\AA}$. How much energy must be supplied to excite the electron from the ground level to the first excited state?
9. An electron is confined to move between two rigid walls separated by $20\text{\AA}$. Find the de Broglie wavelength representing the first three allowed energy states of the electron and the corresponding energies (assume the potential to be zero).
10. A particle is moving in a one-dimensional box (of infinite height) of width $10\text{\AA}$. Calculate the probability of finding the particle within an interval of $1\text{\AA}$ at the centre of the box, when it is in its state of least energy.

UNIT II: Crystallography and X-rays

1. What is a unit cell? How many and which parameters are needed to define a unit cell?
(3 or 6 marks)
2. What is a space lattice? Name the seven crystal systems and mention their characteristic parameters. (4 or 10 marks)
3. What are Bravais lattices? Mention the Bravais lattices possible for different crystal systems.
(3 or 6 marks)
4. What are Miller indices? Mention their salient features. (3 or 6 marks)
5. Explain how the Miller indices of a plane can be determined. (3 or 6 marks)
6. What is inter planar distance? Obtain an expression in terms of lattice parameter and miller indices for the case of a cubic crystal. (4 or 10 marks)
7. What is 'coordination number'? Find the same for simple cubic and body centred lattices.
(3 or 6 marks)
8. What is atomic packing factor? Determine the atomic packing factor for the case of simple cubic, body centred and face centred cubic lattices by calculating number of atoms/unit cell and atomic radius. (4 or 10 marks)
9. Describe the crystal structure of sodium chloride. Write the structural features. (3 or 6 marks)
10. Show that the crystal structure of caesium chloride is simple cubic. Write down the positional coordinates of all ions in the unit cell. Show that there is one molecule per unit cell in its structure. (3 or 6 marks)
11. Describe the crystal structure of zinc sulphide. Give the positional coordinates of sulphur and zinc ions. Show that there are four molecules per unit cell. (3 or 6 marks)
12. What are X-rays? Mention its properties and applications. (3 or 6 marks)
13. What are X-rays? With the neat diagram of an X-ray tube explain the production of X-rays.
14. Explain the origin of continuous X ray spectrum. (4 or 10 marks)
15. Explain the origin of characteristic X ray spectrum. (4 or 10 marks)
16. Derive Bragg's law for the diffraction of X-rays by crystal planes. (3 or 6 marks)
17. Describe the measurement of structural parameters of a crystal using Bragg spectrometer.
(3 or 6 marks)

Related Problems: (3 or 4 marks)

1. Draw a cubic unit cell and plot the following planes: $(1\ 1\ 0)$, $(1\ 0\ 0)$, $(1\ 0\ 1)$, $(2\ 1\ 2)$, $(2\ 2\ 2)$, $(3\ 2\ 1)$, $(\bar{1}\ 1\ 1)$.
2. Find the miller indices of a set of parallel planes which make intercepts in the ratio $a:2b$ on X and Y axes and are parallel to z-axis, where a, b, and c being primitive vectors of the lattice.

3. Find the inter planar spacing with a lattice constant of 0.5nm between the following sets of planes (1 1 0), (1 1 1), (2 1 1).
4. Copper has FCC structure of atomic radius 0.1278nm. Calculate the inter planar spacing for (3 2 1) plane.
5. Cesium crystallizes in a certain type of cubic structure with lattice constant 6.14 Å. Identify the exact type of crystal structure in which it crystallizes, given that the atomic weight and density of cesium are 132.91 and 1900 kg/m³ respectively.
6. A monochromatic X ray beam of wavelength 0.7 Å undergoes first order Bragg reflection from the plane (3 0 2) of a cubic crystal at a glancing angle of 35°. Calculate the lattice constant.

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UNIT III: Conductors and Superconductors

1. What are the basic assumptions of the classical free electron theory? (3 or 6 marks)
2. Obtain an expression for the electrical conductivity of a metal based on the classical free electron theory. (4 or 10 marks)
3. What are the drawbacks (failure) of classical free electron theory? (3 or 6 marks)
4. Mention the important differences between classical free electron theory and quantum free electron theory. (3 or 6 marks)
5. Explain the term drift velocity in metals. (3 or 6 marks)
6. Distinguish between relaxation time and mean collision time. (3 or 6 marks)
7. Mention the various reasons for the scattering of electrons in a metal and their contribution to the electrical resistivity of metals. (3 or 6 marks)
8. Explain the effect of impurities and temperature on the electrical resistivity of metals. (3 or 6 marks)
9. Define the terms 'Fermi factor' and 'Fermi energy'. (3 or 6 marks)
10. Explain the Fermi Dirac distribution of electrons with respect to different temperature conditions. (3 or 6 marks)
11. What is a fermi factor? Explain the fermi factor for different temperature conditions. (3 or 6 marks)
12. What are superconductors? Explain its characteristic properties. (4 or 10 marks)
13. Explain the classifications of superconductors. (3 or 6 marks)
14. Write a note on BCS theory.(3 or 6 marks)
15. Write a note various applications of superconductors. (3 or 6 marks)

Related Problems: (3 or 4 marks)

1. Calculate the drift velocity and thermal velocity of conduction electrons in copper at a temperature of 300K, when a copper wire of length 2m and resistance 0.02Ω carries a current of 15A. Given the mobility of free electrons in copper is $4.3 \times 10^{-3} \text{ m}^2/\text{V}\cdot\text{s}$. (3 or 4 marks)
2. Find the temperature at which there is 2% probability that a state with an energy 0.3 eV above Fermi energy is occupied. (3 or 4 marks)
3. Find the temperature at which there is 1% probability that a state with an energy 0.2 eV above Fermi energy is occupied. (3 or 4 marks)
4. A uniform silver wire has a resistivity of $1.54 \times 10^{-8} \Omega\text{m}$ at room temperature. For an electric field along the wire of 1 volt/cm, compute the drift velocity of an electron and the mobility assuming that there is 5.8×10^{28} conduction electrons/ m^3 . (3 or 4 marks).
5. The critical temperature and critical field for superconducting lead are 7.2K and 800 Gauss respectively. What will be the temperature upto which lead will be in superconducting state in a magnetic field of 400 Gauss?
6. A sample of silicon semiconductor is doped with 10^{22} phosphorous atoms. Calculate its conductivity if mobility of electrons is $0.07 \text{ m}^2/\text{V}\cdot\text{s}$. What is the Hall voltage if this semiconductor with a thickness of $100\mu\text{m}$ and carrying a current of 1 mA is placed perpendicular to a magnetic field of 0.1T.?

UNIT – IV: Semiconductors

1. What is a semiconductor? Distinguish intrinsic and extrinsic semiconductor. (3 or 6 marks).
2. What are direct and indirect band-gap semiconductors? Explain. (3 or 6 marks).
3. What is an intrinsic semiconductor? Explain the mechanism of carrier generation in intrinsic semiconductors. (4 or 10 marks).
4. What are intrinsic semiconductors? Obtain an expression for the conductivity of an intrinsic semiconductor. (3 or 6 marks).
5. Explain the effect of temperature on the conductivity of an intrinsic semiconductor. (4 or 10 marks).
6. What are extrinsic semiconductors? Describe the mechanisms of carrier generation in extrinsic semiconductors. (4 or 10 marks).
7. Obtain an expression for the conductivity of an extrinsic semiconductor. (4 or 10 marks).
8. Explain the effect of temperature on the conductivity of an extrinsic semiconductor. (3 or 6 marks).
9. Explain the effect of temperature on the Fermi level in an intrinsic and extrinsic semiconductor. (4 or 10 marks).

10. Write any four differences between conductors and semiconductors. Increase in temperature decreases the resistivity of a semiconductor, why? (3 or 6 marks).
11. State Hall Effect? Obtain an expression for the Hall voltage and Hall coefficient of an n-type semiconductor. (4 or 10 marks). (3 or 4 marks).
12. Describe the experimental method of determining the carrier concentration in an extrinsic semiconductor using Hall effect. (4 or 10 marks). (3 or 4 marks).
13. What is a P-n junction? Explain its formation. (3 or 6 marks).
14. What is biased and unbiased junction? Explain. (3 or 6 marks).
15. Write a note on breakdown in p-n junction. (3 or 6 marks).
16. What is a zener diode? Explain its V-I characteristics. (3 or 6 marks).
17. Explain the principle, construction and working of light emitting diode. (3 or 6 marks).
18. Explain the principle, construction and working of photo diode. (3 or 6 marks).
19. Write a note on solar cell. (3 or 6 marks).

Related Problems: (3 or 4 marks)

1. The intrinsic carrier density is $1.5 \times 10^{16} \text{ m}^{-3}$ if the electron and hole mobilities are and $0.05 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$, calculate the electrical conductivity of the intrinsic Semiconductor.
2. Mobilities of electrons and holes in a sample of intrinsic germanium at 300K are $0.34 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$ and $0.17 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$ respectively. If the resistivity of the specimen is $2.14 \Omega \text{m}$, compute the intrinsic carrier density.
3. The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3/\text{C}$. The resistivity of the specimen is $8.93 \times 10^{-3} \Omega \text{m}$. Find the mobility and density of the charge carriers.

UNIT V: LASERS and Optical Fibers

1. What is luminescence? Mention the various methods by which luminescence can be achieved. (3 or 6 marks).
2. What is a LASER? Explain its properties. (3 or 6 marks).
3. Distinguish between spontaneous emission and stimulated emission. (3 or 6 marks).
4. Explain how Einstein's theory predicts existence of stimulated emission. (3 or 6 marks).
5. Define the terms 'Metastable state' and 'Population inversion'. (3 or 6 marks).
6. Mention the basic conditions for laser action. How can they be materialised? (3 or 6 marks).
7. What are the basic components required for the construction of a laser? Explain the method of production of laser. (4 or 10 marks).
8. Discuss the relative advantages and disadvantages of three level and four level lasers. (3 or 6 marks).
9. What are the different methods of achieving population inversion? (3 or 6 marks).

10. Describe the construction and working of a Ruby laser. What are its disadvantages? (4 or 10 marks).
11. Describe the construction and working of a He-Ne laser. Mention its applications. (4 or 10 marks).
12. Describe the construction and working of a semiconductor laser. (4 or 10 marks).
13. Briefly mention the applications of lasers. (3 or 6 marks).
14. What is an optical fiber? What is the principle on the basis of which optical transmission is achieved through a fiber? Explain. (3 or 6 marks).
15. Explain the terms numerical aperture, semi angle of the cone of acceptance and relative refractive index difference. (3 or 6 marks).
16. Derive an expression for numerical aperture of an optical fiber in terms of core and cladding. (4 or 10 marks).
17. What are the different modes of propagation possible in optical fibers. Explain in detail with necessary diagrams. (4 or 10 marks).
18. Write any four difference between single mode and multi-mode fibers. (3 or 4 marks).
19. What is attenuation in an optical fiber? Mention the possible reasons for the same. (3 or 6 marks).
20. Describe how the optical fiber can be used in communication. Mention its advantages. (3 or 6 marks).

Related Problems: (3 or 4 marks)

21. A ruby laser pulse duration is 12ps and a power of 200mW. Calculate the number of photons in each pulse, if the laser wavelength is 1060nm. (3 or 4 marks).
22. A He-Ne laser emits light at a wavelength of 632.8 nm and has an output power of 2.3 mW. How many photons are emitted in each minute by this laser when operating? (3 or 4 marks).
23. The ratio of population of two energy level is 1.059×10^{-30} . Calculate the wavelength of the emitted photon at 300 K. (3 or 4 marks).
24. Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV. (3 or 4 marks).
25. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33 (3 or 4 marks).
26. A fiber has a core diameter of 6 μm and its core refractive index is 1.47 and for cladding it is 1.43. How many modes can propagate into the fiber if the wavelength of the source is 1.5 μm . (3 or 4 marks).
27. The fractional index change of optical fiber and refractive index of core are 0.00515 and 1.533 respectively. Determine the refractive index of cladding. (3 or 4 marks).

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Quantum Computing-Question Bank

1. Write the fundamental postulates of quantum mechanics
2. Write any four differences between classical and quantum computing.
3. What is a Linear vector space? Explain its axioms.
4. Explain Dirac Notation and explain how to represents Ket vector and Bra Vector in matrix form.
5. Explain Conjugate of a matrix and Transpose of a matrix.
6. Explain Pauli matrices and the interaction of Pauli matrices on 0 and 1 state.
7. Explain the interaction of Identity operator on 0 and 1 states.
8. Define an observable in quantum mechanics and give an example.
9. Explain how operators are used to represent observables in quantum mechanics.
10. Define eigenvalue and eigenvector in the context of linear algebra.
11. Explain how eigenvalues and eigenvectors are related to observables in quantum mechanics.
12. Define a Hermitian operator and explain its significance in quantum mechanics.
13. Explain how to calculate the expectation value of an observable using matrix representation.
14. Define a unitary operator and explain its role in quantum gates.
15. Show that the product of two unitary operators is also unitary.
16. What are Dirac notations and why is it useful in quantum mechanics?
17. Explain inner product with the bra-ket notations for two quantum states.
18. Explain the normalization and orthogonality properties of quantum states.
19. Define adjoint of an operator.
20. How are quantum states represented as column vectors in matrix notation?
21. Introduce the Pauli matrices and provide their matrix representations.
22. Explain Qubit and its properties
23. What is a qubit? Write any four properties of qubits.
24. Explain Single qubit and Two qubit
25. Explain Quantum superposition, Normalization, and Orthogonality of quantum states
26. Explain unitary Matrix and Pauli matrix
27. Write a note on Bloch Sphere with diagram.
28. Show $|\psi\rangle = |0\rangle$ and $|\psi\rangle = |1\rangle$ on the Bloch sphere.
29. Compare classical and Quantum gates
30. What are quantum gates and how are they different from classical logic gates?
31. Explain Hadamard Gate
32. Give an example of a quantum gate and explain its action on a qubit.
33. Explain Phase Gate (or S Gate) and T Gate.
34. Explain the Pauli X, Y, and Z gates with truth table

35. Explain the operation of a CNOT gate ?
36. Show how the Pauli matrices are related to basic quantum gates like the Hadamard and NOT gates.
37. Represent the qubits in matrix form.
38. Explain how to represent qubits in Bloch sphere.
39. Represent qubits and superposition states in Bloch sphere.
40. Explain the properties of quantum computing.
41. Apply identity Gate on quantum states 0 and 1 to show that there is no change.
42. Explain the quantum NOT gate. How it operates on qubits?
43. Explain Pauli-X gate. How it operates on qubits? Write the truth table for it.
44. Explain Pauli-Y gate. How it operates on qubits? Write the truth table for it.
45. Explain Pauli-Z gate. How it operates on qubits? Write the truth table for it.
46. Use matrix form of H to demonstrate how Hadamard gate operate on qubits. Write the truth table for it.
47. Explain S gate. How it operates on qubits? Write the truth table for it.
48. Explain T gate. How it operates on qubits? Write the truth table for it.
49. what operation does S gate perform on $\alpha|0\rangle + \beta|1\rangle$?

Numericals:

1. Verify if $|w\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $|v\rangle = \begin{pmatrix} 1 & -1 \end{pmatrix}$ are orthogonal.
2. $|\psi\rangle = A(2|0\rangle + 3i|1\rangle)$ find $\langle\psi|\psi\rangle$.
3. Show that (1,-1), (1,2) and (2,1) are linearly dependent.
4. Check for orthogonality $|i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$, $|-i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$.
5. Find if the following matrix is unitary:

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

6. Show that matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$ is unitary

$$7. |\psi\rangle = \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} 2 \\ -i \\ 2-3i \end{pmatrix}$$

find the inner product $\langle\phi|\psi\rangle$

8. Convert Ket vector to Bra vector:

$$|\Psi\rangle = \begin{pmatrix} 2-i \\ 3+2i \end{pmatrix}$$

9. Find Adjoint of Operator A

$$A = \begin{pmatrix} 2 & -i & 4 \\ 1-2i & 5 & 2-5i \\ -5i & 2+i & 3+7i \end{pmatrix}$$

10. Show that $A = \begin{bmatrix} 1 & 4+3i \\ 4-3i & 5 \end{bmatrix}$ is Hermitian